CARTESIAN OBJECTS OF THE DERIVED CATEGORY CHARACTERIZED BY THE SHRIEK-TENSOR MAPS G_{N!}K_N -> G_{N!}Z TENSOR K

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-spaces-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the final equalities in the displayed restrictions (f_varphi^*O_n tensor_{O_m} K_m = K_m, source lines 2806-2809) use that for a morphism of ringed topoi the pullback of the structure sheaf is the target structure sheaf (f_varphi^*O_n = O_m), as set up in the lemma statement; the 'Thus the statement is clear' lines close arguments whose entire componentwise content is displayed immediately above; part (2) additionally needs the flatness assumption to keep the derived tensor termwise on components, as stated.

Editorial note. The label lemma-derived-cartesian-shriek names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Exactness of the component shrieks. Same chapter, \label{lemma-exactness-g-shriek-modules} and the exactness part of \label{lemma-restriction-to-components-site} (card L46149). Lines 2776-2778 and 2791-2794: g_{n!} exact induces a derived functor adjoint to g_n^{-1} resp. g_n^*.
- The component formula for g_{n!}. Same chapter, \label{lemma-restriction-to-components-site} (card L46149) and \label{lemma-restriction-module-to-components-site} (card L46168 of this batch). Lines 2782-2787 and 2799-2809: the restriction to C_m of g_{n!}K_n and of the tensor product is computed by the direct sums over varphi : [n] -> [m].
- Derived tensor product and its restriction to components. Standard interaction of derived tensor with exact functors and restrictions. Lines 2799-2809: the restriction of g_{n!}O_n tensor^L_O K to C_m is the termwise tensor over O_m.
- The unit map of the (g_{n!}, g_n^{-1}) adjunction. Standard adjunction unit for the shriek-pushforward pair. Lines 2778-2782: the canonical map is the adjoint of Z -> g_n^{-1}g_{n!}Z adjoint to id_{g_{n!}Z}.

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Source reference. Original reference 0D74: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 1533–1533. Prerequisite pointer invoked at proof line(s) 2795; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0D72: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 1459–1459. Prerequisite pointer invoked at proof line(s) 2803; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09WG: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 970–970. Prerequisite pointer invoked at proof line(s) 2783; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09WE: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 888–888. Prerequisite pointer invoked at proof line(s) 2753; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In Situation 09WE.*

(1) *An object K of $D(\mathcal{C}_{total})$ is cartesian if and only the canonical map*
$$g_{n!}K_n \longrightarrow g_{n!}\mathbf{Z} \otimes^{\mathbf{L}}_{\mathbf{Z}} K$$
is an isomorphism for all n.

(2) *Let $\mathcal{O}$ be a sheaf of rings on $\mathcal{C}_{total}$ such that the morphisms $f_\varphi^{-1}\mathcal{O}_n \to \mathcal{O}_m$ are flat for all $\varphi : [n] \to [m]$. Then an object K of $D(\mathcal{O})$ is cartesian if and only if the canonical map*
$$g_{n!}K_n \longrightarrow g_{n!}\mathcal{O}_n \otimes^{\mathbf{L}}_{\mathcal{O}} K$$
is an isomorphism for all n.

Proof. Proof of (1). Since $g_{n!}$ is exact, it induces a functor on derived categories adjoint to g_n^{-1}. The map is the adjoint of the map $K_n \to (g_n^{-1}g_{n!}\mathbf{Z}) \otimes^{\mathbf{L}}_{\mathbf{Z}} K_n$ corresponding to $\mathbf{Z} \to g_n^{-1}g_{n!}\mathbf{Z}$ which in turn is adjoint to $\mathrm{id} : g_{n!}\mathbf{Z} \to g_{n!}\mathbf{Z}$. Using the description of $g_{n!}$ given in Lemma 09WG we see that the restriction to $\mathcal{C}_m$ of this map is
$$\bigoplus\nolimits_{\varphi:[n]\to[m]} f_\varphi^{-1}K_n \longrightarrow \bigoplus\nolimits_{\varphi:[n]\to[m]} K_m$$
Thus the statement is clear.

Proof of (2). Since $g_{n!}$ is exact (Lemma 0D74), it induces a functor on derived categories adjoint to g_n^* (also exact). The map is the adjoint of the map $K_n \to (g_n^*g_{n!}\mathcal{O}_n) \otimes^{\mathbf{L}}_{\mathcal{O}_n} K_n$ corresponding to $\mathcal{O}_n \to g_n^*g_{n!}\mathcal{O}_n$ which in turn is adjoint to $\mathrm{id} : g_{n!}\mathcal{O}_n \to g_{n!}\mathcal{O}_n$. Using the description of $g_{n!}$ given in Lemma 0D72 we see that the restriction to $\mathcal{C}_m$ of this map is
$$\bigoplus\nolimits_{\varphi:[n]\to[m]} f_\varphi^*K_n \longrightarrow \bigoplus\nolimits_{\varphi:[n]\to[m]} f_\varphi^*\mathcal{O}_n \otimes_{\mathcal{O}_m} K_m = \bigoplus\nolimits_{\varphi:[n]\to[m]} K_m$$
Thus the statement is clear. □